

Reflection

Before starting my internship, my expectations were fairly low because I initially viewed it as a last resort for gaining industry experience. I was mainly hoping to develop practical engineering skills, experience a professional work environment and leave with something meaningful to add to my experience before graduating. In reality, the internship ended up being far more valuable than I expected. Rather than completing isolated training exercises, I worked as part of a team on a project for Royal North Shore Hospital, helping develop a portable respiratory-testing prototype intended to provide similar functionality to a non-portable system valued at around AU\$70,000. This gave me practical experience with embedded software development, ESP32-based systems and multidisciplinary teamwork while contributing to a prototype with a build cost of under AU\$300.

My main contribution was on the embedded software side of the project. I worked on the ESP32 firmware architecture and developed finite state machine-based logic to manage touchscreen inputs, screen transitions and the respiratory-test workflow. I also worked with sensor acquisition, data logging and system testing. Developing the state machine was a useful example of how different engineering decisions affect the wider system. A touchscreen action could not be treated as an isolated interface change because it also had to fit correctly within the test sequence and work with the sensor and data-handling parts of the prototype. This was the point where I began thinking less about whether my individual software worked and more about how it interacted with the complete engineering system.

One of the biggest lessons I took from the internship was that engineering in industry is not just about whether your individual piece of code or hardware works. It also needs to integrate reliably with the rest of the system and be understandable to the other people working on it. Testing became particularly important because something that worked independently could behave differently once it was integrated with other parts of the prototype. When the solution was not immediately obvious, I had to investigate the problem, discuss possible approaches with the team and iterate until we reached a workable outcome. This made me more comfortable with the uncertainty of real engineering projects, where there is not always a single correct answer.

Working on the ESP32 firmware, touchscreen interface, sensor acquisition and system testing showed me that one of my strengths is working across the hardware and software sides of mechatronics while keeping the wider system in mind. I can contribute technically while communicating with others, adapting to changing requirements and considering how my work affects other parts of a project. My Optik experience and university projects in robotics and embedded systems have given me opportunities to demonstrate these skills.

Most importantly, the internship helped clarify the type of engineering work I want to pursue. Although my main responsibility was embedded software, I found that I was most interested in understanding how the firmware interacted with the electronics, sensors, interface and wider prototype. Seeing those separate parts come together into a working system reinforced my interest in robotics and autonomous systems, particularly roles involving embedded software and systems integration. After graduating, I want to continue developing both my technical depth and my ability to contribute to complete multidisciplinary engineering systems.